Class 5: Data Viz with ggplot

Cecilia Wang (PID:18625854)

Table of contents

Background 1
Gene expression plot 5
Going Further 8
Custom Plots 11

Background

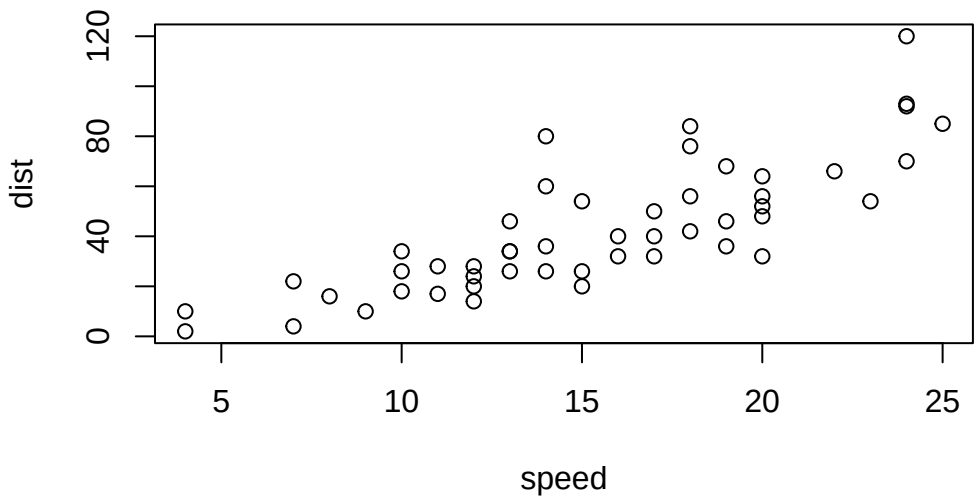
There are a lot of ways to make figures in R. These include so-called "base R" graphics. (e.g. `plot()`) and tones of add-on packages like **ggplot2**.

For example here we make a same plot with both:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



First I need to install the package with the command `install.packages()`.

> **N.B.** We never run an install cmd in a quarto code chunk or we will end up re-installing packages many many times- whcih is not we want!

Every time we want to use one of these "add-on" packages we need to load it up in R with`library()` function:

```
library(ggplot2)
ggplot(cars)
```

Every ggplot need at least 3 things:

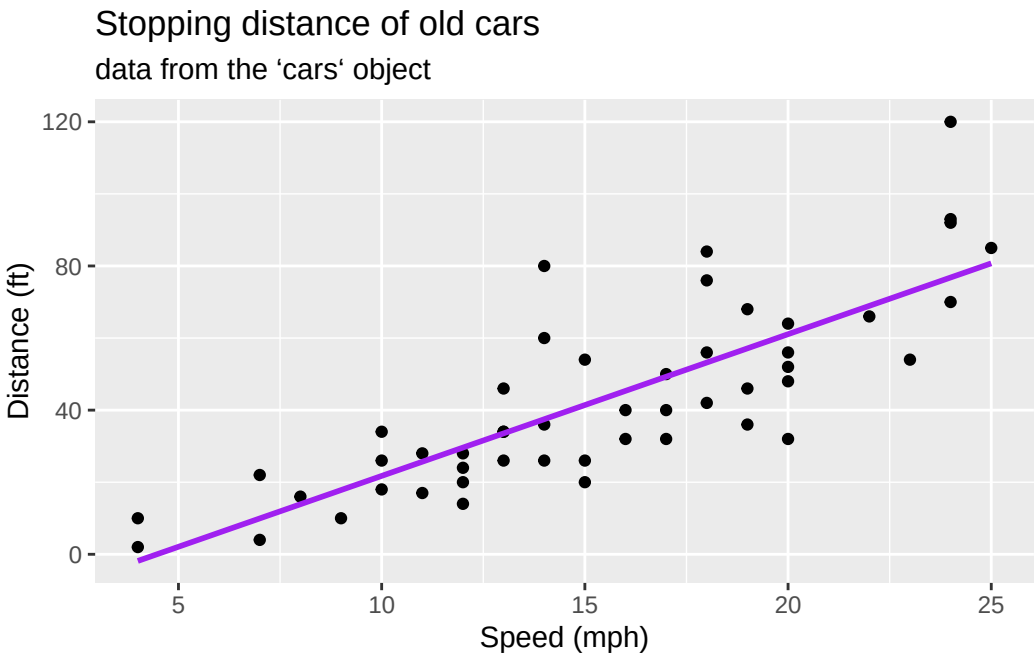
- The **data**, the stuff you want to plotted
- The **aes**thtics, how the data map to the plot
- The **geom**etry, they type of plot

```
p <- ggplot(cars)+
  aes(x=speed, y=dist)+
  geom_point()+
  geom_smooth(method = "lm", se = FALSE, color="purple")+
  labs(title = "Stopping distance of old cars", subtitle = "data from the `cars` object", x="
```

render it out

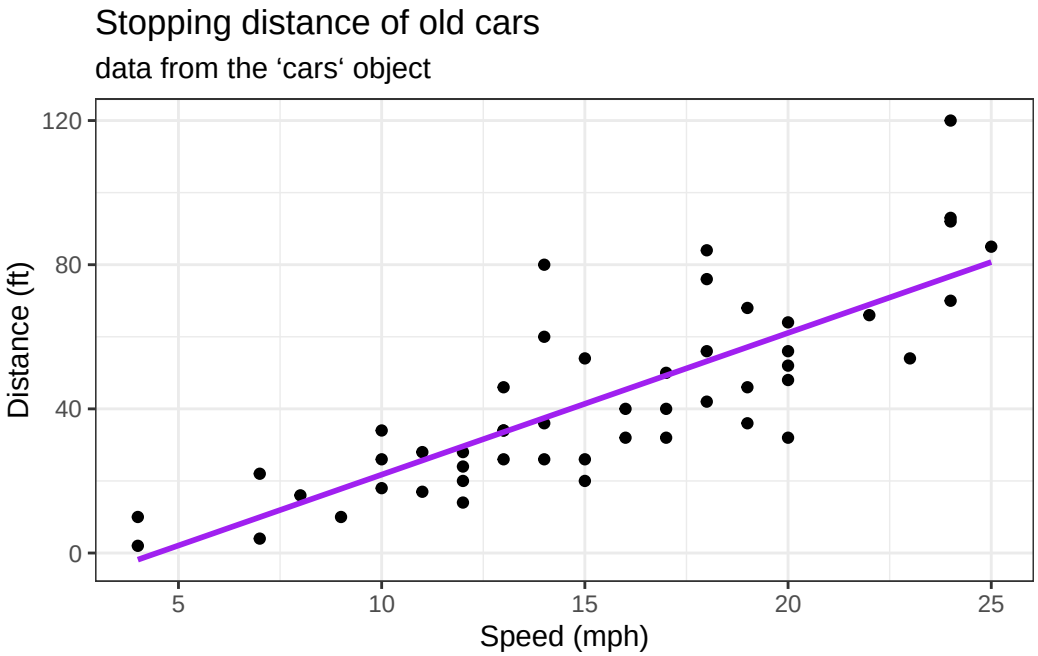
```
p
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
p+theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene expression plot

We can read the input data data the class website

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

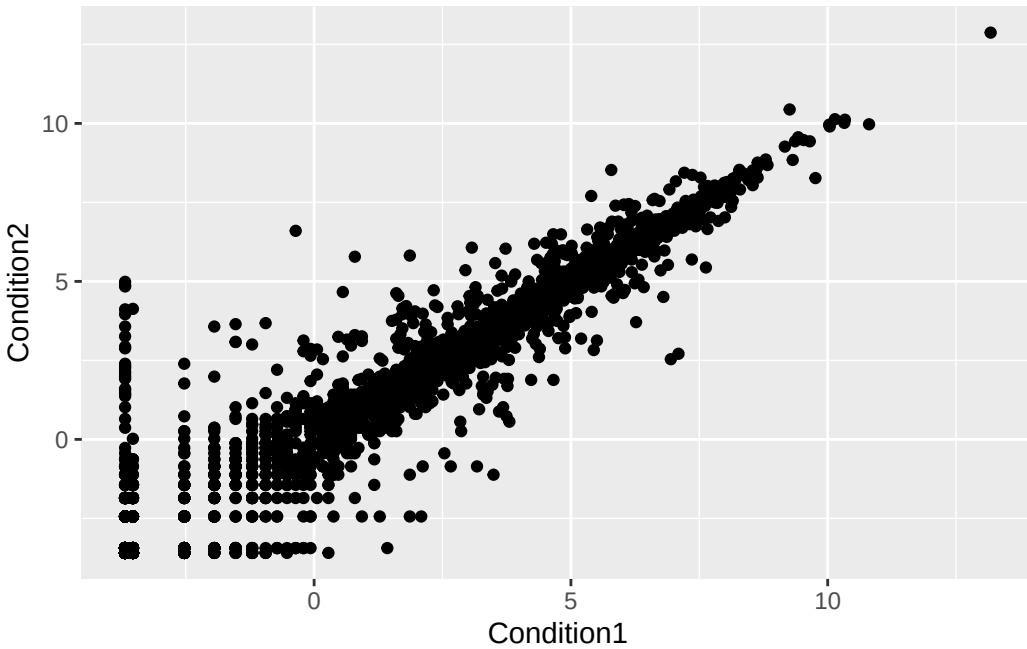
```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
nrow(genes)
```

```
[1] 5196
```

A first version plot

```
ggplot(genes)+
  aes(Condition1, Condition2) +
  geom_point()
```

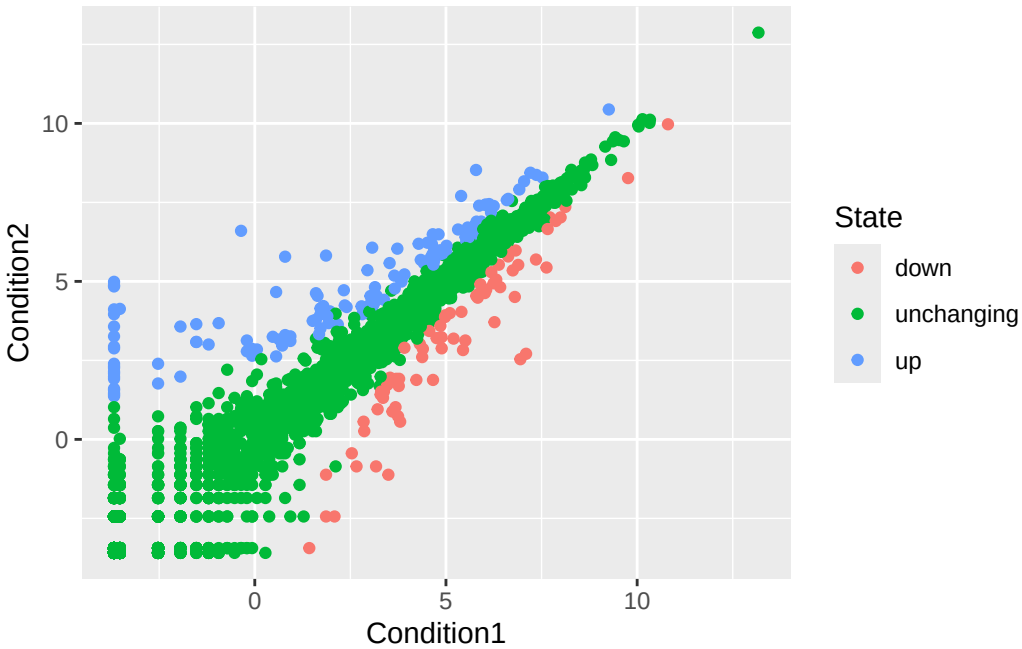



```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

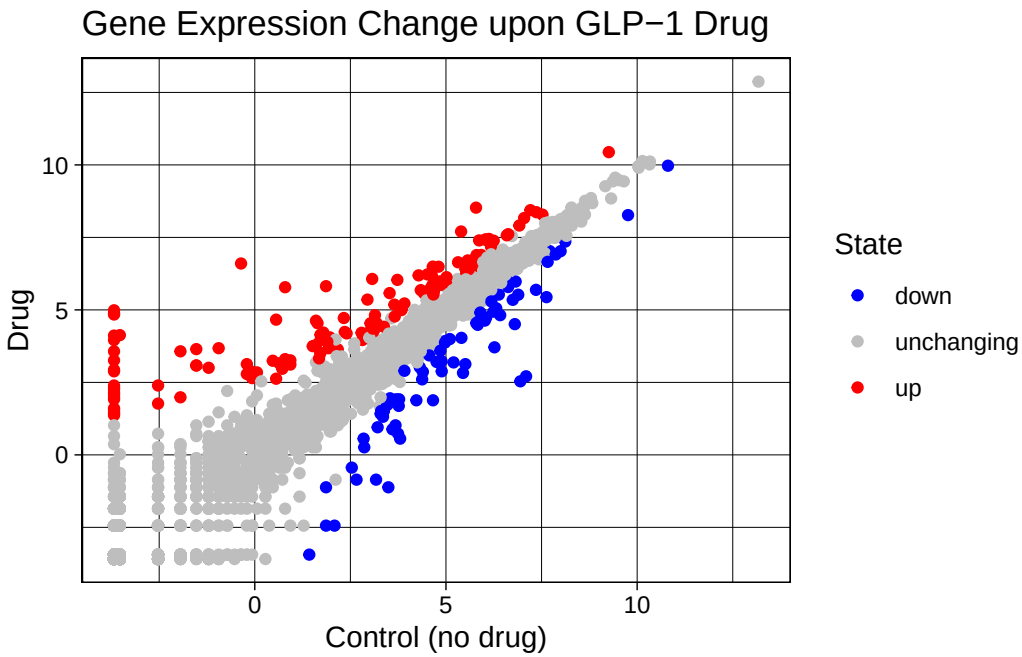
Version 2 plot, let's color by `State` so we can see the "u"p and "down" significant genes compared to all the unchanging genes.

```
ggplot(genes)+
  aes(Condition1, Condition2, col=State) +
  geom_point()
```



Version 3 plot, let's modify the colors to something we like:

```
ggplot(genes)+
  aes(Condition1, Condition2, col=State) +
  geom_point() +
  scale_color_manual(values = c("Blue","grey","red"))+
  labs(x="Control (no drug)", y="Drug",
       title = "Gene Expression Change upon GLP-1 Drug")+
  theme_linedraw()
```



Going Further

Let's have a look at the famous **gapminder**

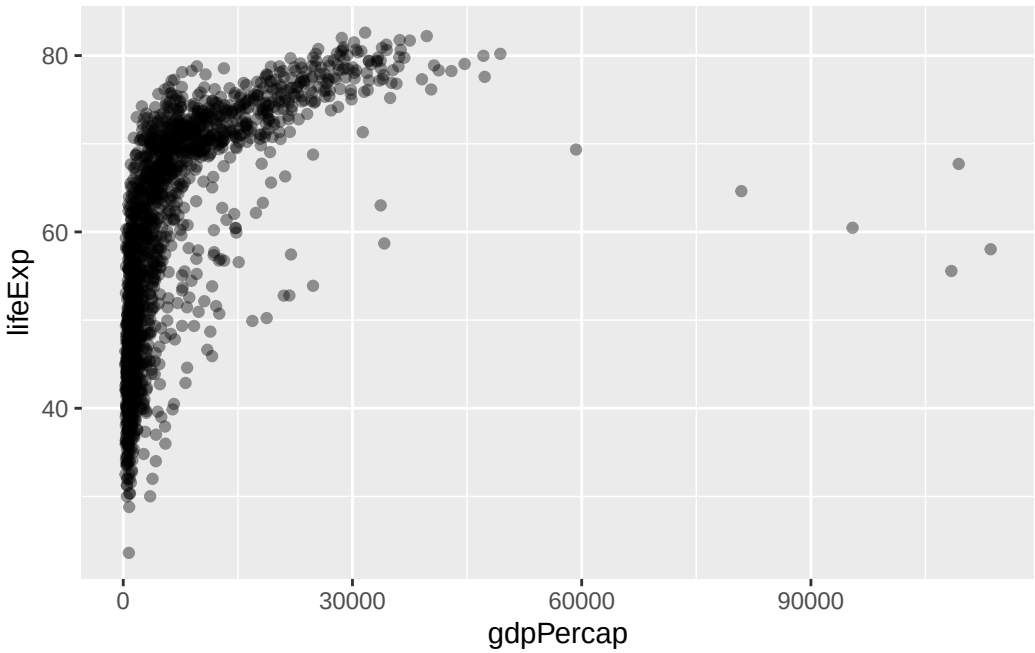
```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv
gapminder <- read.delim(url)
```

```
head(gapminder,5)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
```

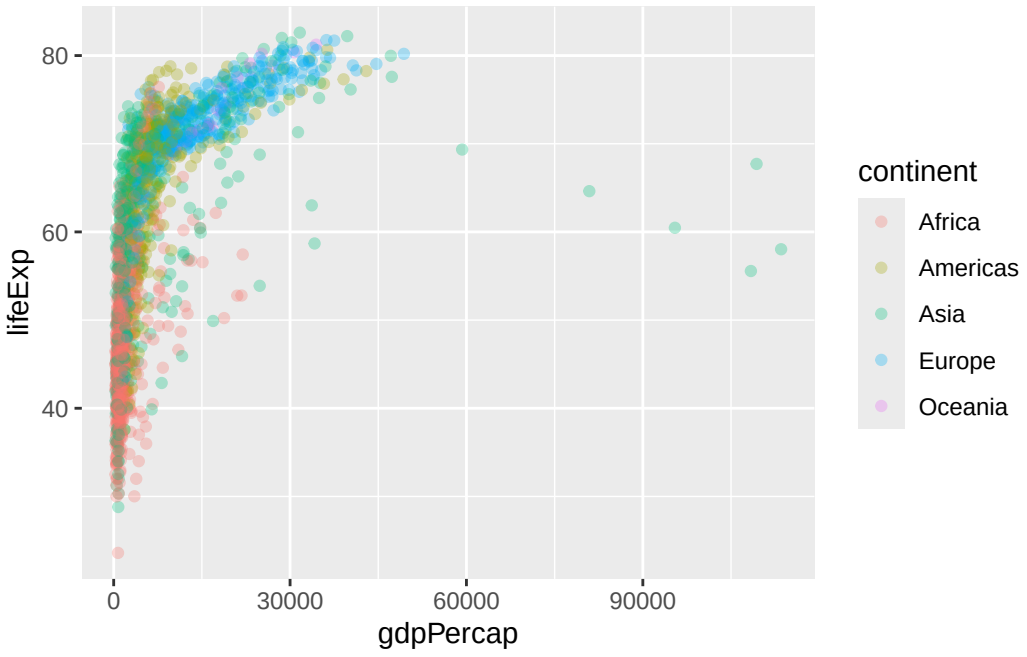
Version 1 Plot

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp)+
  geom_point(alpha=0.4)
```



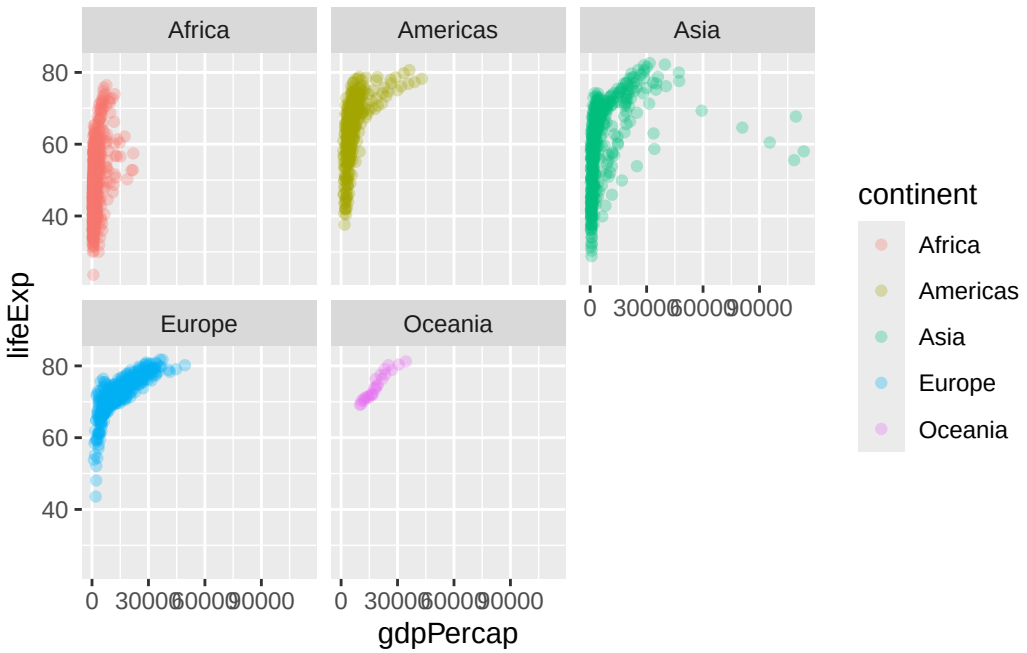
Version 2 plot:

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp,col=continent)+
  geom_point(alpha=0.3)
```



Let's "facet" by continent rather than big hot mess above

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp,col=continent)+
  geom_point(alpha=0.3)+
  facet_wrap(~continent)
```



Custom Plots

How big is this gapminder dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

I want to "filter" down to a subset of this data. I will use **dplyr** package to help me.

First I need to install and load it up…

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
gapminder_1977 <- gapminder %>% filter(year==1977)
head(gapminder_1977)
```

```
      country continent year lifeExp      pop  gdpPercap
1 Afghanistan      Asia 1977  38.438 14880372   786.1134
2     Albania    Europe 1977  68.930  2509048  3533.0039
3     Algeria    Africa 1977  58.014 17152804  4910.4168
4      Angola    Africa 1977  39.483  6162675  3008.6474
5   Argentina  Americas 1977  68.481 26983828 10079.0267
6   Australia   Oceania 1977  73.490 14074100 18334.1975
```

```r
gapminder_2007 <- gapminder %>% filter(year==2007)
head(gapminder_2007)
```

```
      country continent year lifeExp      pop  gdpPercap
1 Afghanistan      Asia 2007  43.828 31889923   974.5803
2     Albania    Europe 2007  76.423  3600523  5937.0295
3     Algeria    Africa 2007  72.301 33333216  6223.3675
4      Angola    Africa 2007  42.731 12420476  4797.2313
5   Argentina  Americas 2007  75.320 40301927 12779.3796
6   Australia   Oceania 2007  81.235 20434176 34435.3674
```

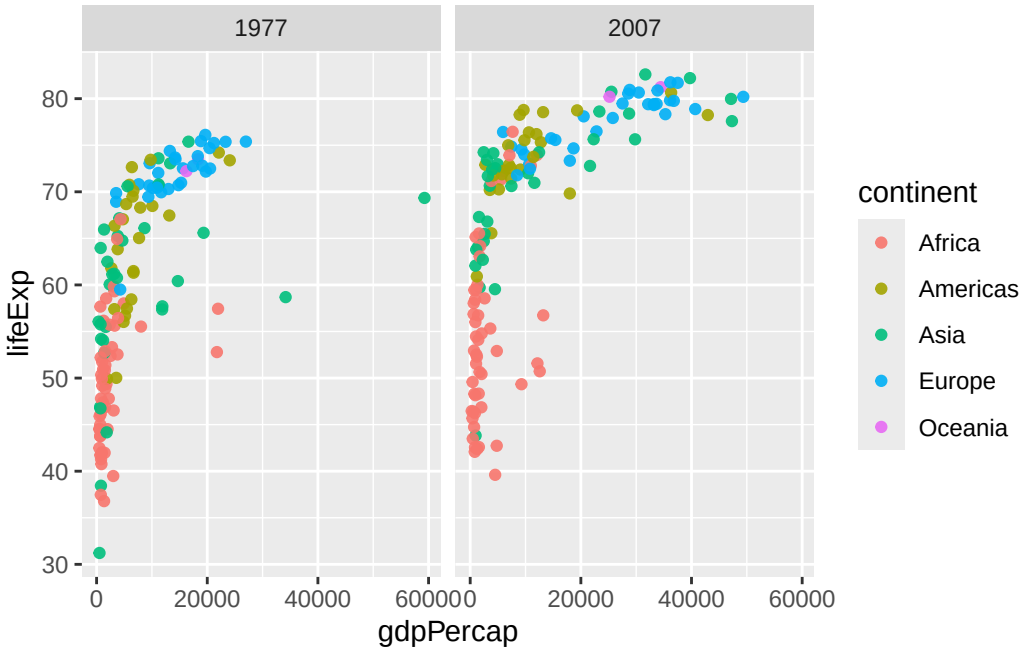
```r
filter(gapminder_2007, country=="United States")
```

```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 2007  78.242 301139947  42951.65
```

Q. Make a plot compring 1977 and 2007 for all countries.

```r
gap_compare <- filter(gapminder,year %in% c(1977,2007))

ggplot(gap_compare)+
  aes(x=gdpPercap, y=lifeExp, col=continent)+
  geom_point(alpha=0.9)+
  facet_wrap(~year)
```


1977
2007
lifeExp
80
70
60
50
40
30
0
20000
40000
60000
0
20000
40000
60000
gdpPercap
continent
Africa
Americas
Asia
Europe
Oceania